Invariant subalgebras of Orlik-Solomon algebras in type A

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Plan

I. MAIN THEOREM: The theoretical minimum

II. Why care?

III. Ideas of proofs

Invariant theory

Let a group G act on a ring R

The invariant (sub) ring is

$$R^G := \{x \in R : g \cdot x = x \ \forall \ g \in G\}.$$

1st and 2nd fundamental theorems

I. A "nice" generating set for R^G

II. Relations among those generators

Example

Let the symmetric group $\mathfrak{S}_n$ act on $\mathbb{C}[\underline{x}] = \mathbb{C}[x_1, \ldots, x_n]$ permuting subscripts

1. $\mathbb{C}[\underline{x}]^{\mathfrak{S}_n}$ is generated by
   - elementary symmetric polynomials $\{e_i\}_{i=1}^n$

     or
   - power sum symmetric polynomials $\{p_i\}_{i=1}^n$
2. $\{e_i\}_{i=1}^n$ is algebraically independent

   $\{p_i\}_{i=1}^n$ is algebraically independent

Example (Chevalley)

Let a rk-n complex reflection group W act on $\mathbb{C}[\underline{x}]$.

1. $\mathbb{C}[\underline{x}]^W$ is generated by n homogeneous polynomials.

2. Those n homogeneous polynomials are algebraically independent.

Example

Main theorems of this talk.

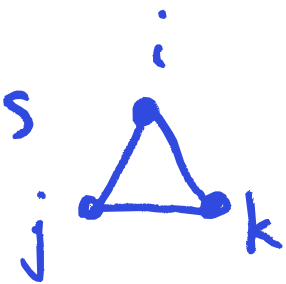
$G = \mathfrak{S}_n \times \mathfrak{S}_1 \leq \mathfrak{S}_{n+1}$ acting on....

- $OS_n = OS(\mathcal{A}_n)$
- $VG_n = VG(\mathcal{A}_n)$

Orlik-Solomon and Varchenko-Gel'fand rings:

$$OS_n := \frac{\Lambda_{\mathbb{Q}}\{e_{ij}\}_{1\le i<j\le n+1}}{\langle e_{ik}e_{jk} - e_{ij}e_{jk} + e_{ij}e_{ik}\rangle}$$

2 edges of



$$VG_n := \frac{\mathbb{Q}[x_{ij}]_{1\le i,j\le n+1}}{\langle x_{ij}^2,\ x_{ij}+x_{ji},\ x_{ik}x_{jk} - x_{ij}x_{jk} + x_{ij}x_{ik}\rangle}$$

$\mathfrak{S}_n \leq \mathfrak{S}_{n+1}$ acts by permuting subscripts.

Example

A $\mathbb{Q}$-basis for OS_2 is

$$\left\{ \begin{array}{lll} 1, & e_{12} & , \; e_{12}e_{13} \\ & e_{13} & \quad\;\; e_{12}e_{23} \\ & e_{23} & \end{array} \right\}$$

$(12)\, e_{13} = e_{23}$ and $(23) \cdot e_{12}e_{13} = -e_{12}e_{13}$

Other exterior monomials:

$$e_{13}e_{23} = e_{12}e_{23} - e_{12}e_{13}$$

$$e_{12}e_{13}e_{23} = e_{12}(e_{12}e_{23} - e_{12}e_{13}) = 0$$

Example

A $\mathbb{Q}$-basis for VG_2 is

$$\left\{ \begin{array}{lll} 1, & x_{12} & , \; x_{12}x_{13} \\ & x_{13} & \quad\;\; x_{12}x_{23} \\ & x_{23} & \end{array} \right\}$$

$(12)\, x_{12} = -x_{12}$ and $(23) \cdot x_{12}x_{13} = x_{12}x_{13}$

So $OS_n \cong VG_n$ as vector spaces but not as rings or representations

Free Graded-Commutative algebra

Let

- A be an alphabet, $|\cdot| : A \to \mathbb{Z}$
- $\mathbb{Q}\langle A \rangle$ free associative algebra

<u>Def</u> The free g.c. algebra is

$$\mathbb{Q}\{A\} := \mathbb{Q}\langle A \rangle / \langle ab - (-1)^{|a||b|} ba \rangle$$

Examples

- $|a| = 2 \longleftrightarrow$ polynomial ring $\mathbb{Q}[A]$

- $|a| = 1 \longleftrightarrow$ exterior algebra $\Lambda_{\mathbb{Q}}\{A\}$

Examples

- $A = \{\alpha, \mu, \gamma\}$ $\quad |\alpha| = |\mu| = 1, \quad |\gamma| =$

  $\mathbb{Q}$-basis:

  $$\left\{1, \begin{matrix}\alpha \\ \mu\end{matrix}, \begin{matrix}\alpha\mu \\ \gamma\end{matrix}, \begin{matrix}\alpha\gamma \\ \mu\gamma\end{matrix}, \begin{matrix}\alpha\mu\gamma^2 \\ \gamma^3\end{matrix}, \dots\right\}$$

  $$\mathrm{Hilb}(\mathbb{Q}\{A\}; t) = \frac{(1+t)^2}{1-t^2} = \frac{1+t}{1-t}$$

Def Let $p = \lfloor \frac{n+1}{2} \rfloor$, the ideal of $\mathbb{Q}\{\alpha, \mu, \gamma\}$

$$I_n = \begin{cases} \langle \gamma^p \rangle & n \text{ even} \\ \langle \gamma^p, \alpha\gamma^{p-1} - (p-1)\mu\gamma^{p-1} \rangle & n \text{ odd} \end{cases}$$

Theorem (K. '26)

$$\overline{OS}_n = OS_n^{\mathfrak{S}_n} = \mathbb{Q}\{\alpha, \mu, \gamma\} / I_n.$$

$\alpha \longleftrightarrow$ orbit sum of $e_{1,2}$

$\mu \longleftrightarrow$ orbit sum of $e_{1,n+1}$

c = orbit sum of $e_{1,2}e_{1,n+1}$

$\gamma \longleftrightarrow \alpha\mu - \rho c$

<u>Theorem</u> (K. '26)

Let $z = \sum_{i=1}^{n} x_{i,n+1} \in VG_n$.

Then

$$\overline{VG}_n := VG_n^{\mathfrak{S}_n} = \mathbb{Q}[z]/(z^{n+1})$$

So we have 1st & 2nd fundamental theorems for

	$\overline{OS}_n$	&	$\overline{VG}_n$
I.	α, μ, γ		z
II.	I_n		$z^{n+1} = 0$

Why care?

Let $\mathrm{Conf}_{n+1}(\mathbb{C})$ denote configuration space of points in the plane.

<u>Famous fact</u> (Arnol'd 1969)

$$OS_n \cong H^\bullet(\mathrm{Conf}_{n+1}(\mathbb{C}); \mathbb{Q})$$

Moreover,

$$VG_n \cong H^\bullet(\mathrm{Conf}_{n+1}(\mathbb{R}^3); \mathbb{Q})$$

Why invariant theory?

If $\pi : X \longrightarrow X/G$ is a covering space then

$$H^\bullet(X/G) = H^\bullet(X)^G$$

Example:

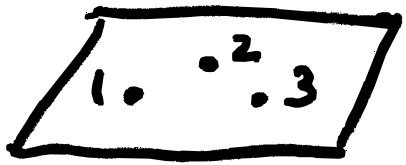
$\mathfrak{S}_{n+1}$ acts freely on $\mathrm{Conf}_{n+1}(\mathbb{C})$

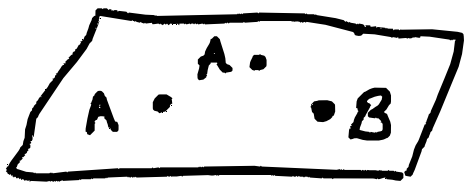
$$\pi : \mathrm{Conf}_{n+1}(\mathbb{C}) \longrightarrow \mathrm{Conf}_{n+1}(\mathbb{C}) / \mathfrak{S}_{n+1}$$

$$\mathrm{Conf}_{n+1}(\mathbb{C}) / \mathfrak{S}_{n+1} = \mathrm{PConf}_{n+1}(\mathbb{C})$$

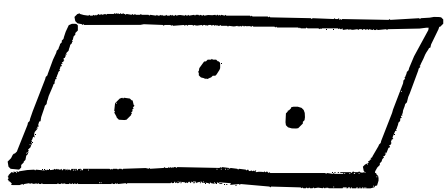
So $H^\bullet(\mathrm{PConf}_{n+1}(\mathbb{C})) = OS_n^{\mathfrak{S}_{n+1}}$

$$= \mathbb{Q}\{\alpha\} / \alpha^2$$

$\mathrm{Conf}_{n+1}(\mathbb{C}) \ni$ 

$\mathrm{MConf}_{n,1}(\mathbb{C}) \ni$ 

$$\mathrm{MConf}_{n,1}(\mathbb{C}) = \mathrm{Conf}_{n+1}(\mathbb{C}) / \mathfrak{S}_n \times \mathfrak{S}_1$$

$\mathrm{PConf}_{n+1}(\mathbb{C}) \ni$ 

UPSHOT:

$$H^\bullet(\mathrm{MConf}_{n,1}(\mathbb{C})) \cong \overline{OS}_n \cong \frac{\mathbb{Q}\{\alpha,\mu,\gamma\}}{I_n}$$

$$H^\bullet(\mathrm{MConf}_{n,1}(\mathbb{R}^3)) \cong \overline{VG}_n \cong \mathbb{Q}[z]/(z^{n+1})$$

Proof Ideas

For $\overline{VG_n}$:

As $\mathfrak{S}_{n+1}$ reps, $VG_n \cong \mathbb{Q}[\mathfrak{S}_{n+1}]$.

$\dim(\overline{VG_n})$ = multiplicity of trivial $\mathfrak{S}_n$ rep. inside of $\mathbb{Q}[\mathfrak{S}_{n+1}]\downarrow_{\mathfrak{S}_n}$

$$= \langle \mathbb{1}_{\mathfrak{S}_n}, \mathbb{Q}[\mathfrak{S}_{n+1}]\downarrow_{\mathfrak{S}_n} \rangle$$

$$= \langle \mathbb{1}_{\mathfrak{S}_n}\uparrow^{\mathfrak{S}_{n+1}}, \mathbb{Q}[\mathfrak{S}_{n+1}] \rangle$$

$$= \langle h_n h_1, h_1^{n+1} \rangle = n+1$$

The basis of $\overline{VG}_n$ consists of products

$$x_{i_1 j_1} x_{i_2 j_2} \cdots x_{i_d j_d}$$

where $|\{i_1, \ldots, i_d\}| = d$.

Thus powers of z are nonzero until the pigeonhole principle forces them to be.

$\{1, z, z^2, \ldots, z^n\}$ are all nonzero and invariant

so

$$\mathbb{Q}[z]/(z^{n+1}) \subseteq \overline{VG}_n$$

and

$$\dim \mathbb{Q}[z]/(z^{n+1}) = \dim \overline{VG}_n.$$

Proof of $\overline{OS_n}$ is trickier, but follows same flow

- Vector space dim $\overline{OS_n} = 2n$
- Hilbert series $= (1+t)(1+t+\cdots+t^{n-1})$
- Linear basis

$$\left\{ \begin{matrix} 1, & \alpha, & \gamma, & \alpha\gamma, & \alpha\mu\gamma, & \ldots, & \alpha\mu\gamma^{p-1} \\ & \mu, & \alpha\mu, & \mu\gamma, & \gamma^2 & & \end{matrix} \right\}$$

- Presentation

Thank You!